Joseph Haas

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Senior Software Engineer | Technical Product Manager

Backend & API Development ▪ Payments & Benefits Systems ▪ Product Strategy & Delivery

Senior Software Engineer with 5 years building production backend systems for payments and ledger platforms that fund, track, and reconcile roughly $1B a year in benefits for 100+ Fortune 1000 employer partners and 3.37M learners, plus 8 years as a product manager shipping enterprise software. I specialize in financial systems where correctness and money are the same problem—API design, distributed architecture, and cross-functional delivery that bridges engineering, product, and operations.

Areas of Expertise:

- *Backend & API Design*
- *Event-Driven Architecture*
- *System Architecture & ADRs*
- *Performance & Load Testing*
- *Product Strategy & Roadmaps*
- *Payments & Ledger Systems*
- *Data Modeling & Database Design*
- *Observability & Alerting*
- *Security Remediation*
- *Agile, BDD & SAFe Delivery*
- *Distributed Systems*
- *Data Migrations & Backfills*
- *Incident Response & Forensics*
- *B2B & Benefits Integration*
- *Technical Documentation & Runbooks*

Technical Skills: Node.js, TypeScript, Python, SQL, PostgreSQL, Snowflake, REST API development, GraphQL (AWS AppSync), event-driven architecture, AWS (Lambda, AppSync, RDS), containerization, Git, GitHub Actions (CI/CD), Mocha/Chai/Sinon, integration testing, BDD, Snyk, Salesforce and NetSuite integrations, Jira, Confluence

Professional Experience

GUILD – Denver, CO 2020 – Present

Education and career-mobility benefits platform serving 100+ Fortune 1000 employers; the payments engineering team owns the systems that fund, track, and reconcile roughly $1B a year in learner benefit dollars across 3.37M learners.

Senior Software Engineer *2024 – Present*

- Led the migration of 100% of the platform (100+ employer partners, 3.37M learners) to a new eligibility system: designed the API contract, authored technical specs, and rewrote tax-treatment and spending-limit logic. Built a production side-by-side parity framework and cut over with no rollbacks and no incidents.
- Built a bulk account-lookup endpoint (up to 500 accounts per call) replacing thousands of individual requests, then refactored it to fetch benefit configuration once per batch, cutting mean latency 55–74% and raising throughput 2–4× for batches of 50–500 accounts in production. Automated the load tests that measured it.
- Led research and adoption of event-driven integration for secure automated partner data ingestion. Designed the event schema and authored the integration rollout plan.
- Built core features for partner calendar organization system: bidirectional sync, gap-tolerant ruleset engine, and integration test infrastructure.
- Hardened service availability: built metric-based alerting and runbooks to detect spend-check failures, automated deploy validation with scripted smoke tests, and remediated critical vulnerabilities.
- Led incident investigations and data forensics across our team's service and Snowflake, turning each into a reusable runbook.

Software Engineer II *2022 – 2024*

- Architected and shipped createSpend, the core spend-writing API behind roughly $1B a year across four benefits, with complex funding determination, idempotency, and partner-aware writes—retiring fragmented legacy patterns and unifying data modeling.
- Owned the validation and benefit utilization correction service from inception, building the endpoints that carry internal identifiers, external partner identifiers, and eligibility context into every spend decision in systems.
- Built external reporting system end-to-end (backend API + frontend) enabling self-service reporting for B2B partners.
- Built new core platform foundation: auth scopes, read replicas, bulk update endpoint, and event-driven consumer. Dual-wrote adjustment and data-merge events for ledger-accurate spend tracking; authored the backfill and migration communication strategy.
- Integrated 2 core platforms, removing a cross-team dependency and simplifying data resolution while reducing downtime.
- Authored the team's ADRs and architecture docs for four core features and the incident runbook.

Software Engineer I *2021 – 2022*

- Delivered the Salesforce reconciliation integration, enabling accurate end-of-year benefit accounting and preventing employer over- and underpayment across fiscal-year boundaries.
- Extended the funding API with filtering, pagination, bulk adjustments, and tax-period endpoints, and integrated it into the company's GraphQL gateway.
- Built an internal admin tool for bulk spend corrections so Operations could fix data without engineering involvement.
- Onboarded nine employer partners through benefit configuration and SQL data migrations, wrote runbooks that shortened every launch after, and migrated security scanning from ShiftLeft to Snyk across five repositories.

Product Manager, Payment Products *2020 – 2021*

- Owned balance tracking and external benefits administration products on the cross-functional payments scrum teams, serving Fortune 1000 employer partners and the systems that determine how each learner's benefit dollars are funded and reconciled.
- Wrote detailed user stories and front-end mockups, defined MVPs, and prioritized the backlog for engineering teams delivering payments features on a two-week cadence.
- Coordinated multi-team projects with fellow product managers and engineering leaders, aligning requirements, deployment sequencing, and cross-squad dependencies for releases spanning several services.

MJ FREEWAY – Denver, CO 2018 – 2019

Software Product Manager, Core Services

- Product owner for the Core Services suite (Cultivation, Processing, Point of Sale); defined the end-to-end enterprise development strategy on the SAFe framework and kept releases shipping every two to three weeks.
- Wrote requirements, user stories, wireframes, and UI mockups; collaborated with enterprise clients to validate roadmap milestones and reduce refactors.
- Trained the product organization on SDLC practices, from bug and story writing to identifying core customer needs; led the cross-functional evaluation of new global markets.

CHARTER COMMUNICATIONS – Denver, CO 2014 – 2018

Manager, Video Products

- Defined the roadmap and owned the backlog for three TiVo set-top video guide products supporting 20% of enterprise video customers; managed cross-platform features across set-top box, Roku, Xbox, Samsung, web, and mobile.
- Feature owner for accessibility compliance: partnered with UX to ship a text-to-speech guide usable without vision, plus a pre-installed Samsung tablet app and a web portal with cloud DVR.
- Won the Charter Champion Award (2015) for rebuilding internal and customer-facing support tools with the Care team, reducing call handling time and adding self-service options.

iBAHN – Denver, CO 2012 – 2014

Associate Product Manager, IPTV Platform / Hospitality Public PC Solutions

- Managed the ETVI in-room video platform for hotels (custom set-top hardware, live TV and on-demand UI, room-control integrations) and led marketing, business development, and global product development for a public-PC product for hotel lobbies and business centers.
- Gathered UI, custom-skin, and feature requirements for large clients including Marriott, Hilton, and Four Seasons; turned customer issues into feature requirements and better support and error handling.
- Analyzed market trends and competitor data to identify product gaps; directed an internship team in competitive research against the ETVI product.

Education & Certification

Back-End Engineering Program | Turing School of Software and Design, Remote
Bachelor of Science in International Business, minor in Marketing | University of Denver, Denver, CO
Certifications: Certified Scrum Product Owner (Scrum Alliance) ▪ Pragmatic Marketing Certification (Pragmatic Institute)